In 2019 we recognized other income, net of expenses, of $103 million, increasing compared to $53 million in 2018, mainly benefitting from the grants associated with the programs part of the European Commission IPCEI in Italy and in France, partially offset by a higher level of start-up costs associated with the production ramp up of the 200 mm fab recently acquired from Micron Technology Inc. in Singapore.

In 2018 we recognized other income, net of expenses, of $53 million, slightly decreasing compared to $55 million in 2017, mainly due to lower level of R&D grants.

Impairment, restructuring charges and other related closure costs

	Year Ended December 31,		
	2019	2018	2017
		(In millions)	
Impairment, restructuring charges and other related closure costs	$ (5)	$ (21)	$ (45)

In 2019 we recorded $5 million of impairment, restructuring charges and other related closure costs, mainly consisting of impairment of equipment and licenses dedicated exclusively to certain development projects that were cancelled, while no alternative future use was identified internally.

In 2018 we recorded $21 million of impairment, restructuring charges and other related closure costs, consisting of: (i) $19 million related to the set-top box restructuring plan and (ii) $2 million of impairment of acquired technologies, for which it was determined that they had no future alternative use.

In 2017 we recorded $45 million of impairment, restructuring charges and other related closure costs, primarily consisting of: (i) $34 million of net restructuring charges related to the set-top box restructuring plan; (ii) $13 million of restructuring charges related to the restructuring plan in Bouskoura, Morocco; (iii) $3 million charge relating to the update of the existing unused lease provision and (iv) $5 million income for the reversal of provisions related to previously announced restructuring plans, mainly the Embedded Processing Solutions business restructuring plan, for which accrued provisions were not fully used at completion of the plan.

Operating income

	Year Ended December 31,		
	2019	2018	2017
		(In millions)	
Operating income	$ 1,203	$ 1,400	$ 1,005
As percentage of net revenues	12.6%	14.5%	12.0%

Operating income in 2019 was $1,203 million, decreasing by $197 million compared to 2018, reflecting normal price pressure, increased unsaturation charges and higher R&D spending, partially offset by higher level of grants and favorable currency effects, net of hedging.

Operating income in 2018 was $1,400 million, improved by $395 million compared to 2017, reflecting higher volumes, improved manufacturing efficiencies and product mix and lower restructuring charges, partially offset by unfavorable currency effects, net of hedging, normal price pressure and higher operating expenses.